COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
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ACTIVITY NO. ___ PRELIM EXAM

1. PROCEDURE OUTPUT

Part A: Installing and Verifying Ansible

1. Update the package repository.
2. Install pip if it is not yet installed.
3. Install Ansible.
4. Verify that Ansible is installed correctly.
5. Verify SSH connectivity to each managed node.

Submit a screenshot of the following:

- Ansible version

```
duenas@server1:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/duenas/.ansible/plugins/modules', '/us
r/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/duenas/.ansible/collections:/usr/share/ans
ible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/p
ython3)
  jinja version = 3.1.2
  libyaml = True
```

```
duenas@server2:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/duenas/.ansible/plugins/modules', '/us
r/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/duenas/.ansible/collections:/usr/share/ans
ible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/p
ython3)
  jinja version = 3.1.2
  libyaml = True
```

● SSH connectivity

```
ubuntu2@server2:~$ ssh ubuntu1@192.168.56.108
ubuntu1@192.168.56.108's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Aug 13 01:05:57 2026 from 192.168.56.109
```

```
ubuntu1@server1:~$ ssh ubuntu2@192.168.56.109
ubuntu2@192.168.56.109's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Aug 13 01:05:15 2026 from 192.168.56.108
```

Part B: Building an Inventory File

Inventories provide Ansible with information about managed nodes and their network locations.

1. Create a working directory.
2. Create an inventory file.
3. Add the following content.
4. Verify the inventory structure.
5. Ping all managed nodes.

Demonstrate the following:

● inventory.yaml

```
all:
  children:

    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.108

        ubuntu2:
          ansible_host: 192.168.56.109
```

- Successful inventory verification

```
ubuntu2@server2:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "ubuntu1": {
                "ansible_host": "192.168.56.108"
            },
            "ubuntu2": {
                "ansible_host": "192.168.56.109"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "ubuntu"
        ]
    },
    "ubuntu": {
        "hosts": [
            "ubuntu1",
            "ubuntu2"
        ]
    }
}
```

- Successful ping results

```
ubuntu2@server2:~/ansible-lab$ ansible all -m ping -i inventory.yaml
ubuntu1 | UNREACHABLE! => {
    "changed": false,
    "msg": "Failed to connect to the host via ssh: ubuntu2@192.168.56.108: Permi
ssion denied (publickey,password).",
    "unreachable": true
}
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Part C: Creating a Metagroup

Metagroups allow multiple groups to be combined into a single logical group.

1. Modify the inventory file.
2. Test the metagroup.
3. Display all hosts belonging to the metagroup.

- Demonstrate successful execution of the metagroup.

Part D: Creating Variables

Variables allow information such as operating systems and usernames to be assigned to hosts and groups.

1. Add host variables.
2. Add group variables.
3. Create a playbook to display variables.
4. Insert the following content.
5. Run the playbook.

- Demonstrate successful use of variables.

Part E: Creating Your First Playbook

Playbooks are YAML files used to automate tasks on managed nodes.

1. Create a playbook.
2. Add the following content.
3. Execute the playbook.
4. Observe the results. Successful tasks should display an ok status and a play recap summary.

Demonstrate:

- Successful playbook execution
- Successful task completion on all managed nodes

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a new group named servers that contains only:

- ubuntu1
- centos1

Create a playbook named challenge.yaml that displays the message:
[Welcome to Ansible Laboratory]
Run the playbook and demonstrate that only hosts in the servers group receive the message.

Guide Questions

1. What is the role of the Control Node in Ansible?
   -
2. Why is an inventory file required?
   -
3. What is the purpose of a metagroup?
   -
4. How do variables simplify Ansible administration?
   -
5. What is the difference between an ad hoc command and a playbook?
   -

Conclusion
Today we did the preliminary exam about ansible and ssh. Only the part A did i accomplish the task, i had an error when i did the ping on part B, that's why i haven't finished the rest. Overall I think I can finish this if i only know how to fix the error.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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